Attendance Recovery Computation (Case Study 12)

Problem Statement

Find a student's current attendance percentage and calculate how many consecutive classes the student must attend to
reach a required attendance percentage.

Objectives

Use variables, arithmetic, input, and a while loop.

Assumptions

Classes attended and total classes are non-negative; total classes is greater than zero; the student attends every future
class.

Algorithm

1. Read attended classes, total classes, and target percentage. 2. Calculate current percentage. 3. Add one attended class
and one total class repeatedly until the percentage reaches the target. 4. Print both answers.

Flowchart (Text Form)

START → Read inputs → Calculate current % → Target reached? → If NO, add one attended and total class → Check
again → Print result → STOP.

Pseudocode

```
R
```

Python Source Code

```
attended = int(input("Classes attended: "))
total = int(input("Total classes: "))
target = float(input("Target attendance (%): "))
if total &lt;= 0 or attended &lt; 0 or attended &gt; total:
    print("Invalid attendance data")
else:
    current = attended / total * 100
    extra = 0
    while (attended + extra) / (total + extra) * 100 &lt; target:
        extra += 1
    print("Current attendance:", round(current, 2), "%")
    print("Extra classes to attend:", extra)
```

Test Case and Expected Output

Input: attended=30, total=40, target=75
Expected output: Current attendance: 75.0 %; Extra classes to attend: 0

Conclusion

The program calculates attendance and finds the minimum extra classes needed under the stated assumption.